SQROCK IT SOLUTIONS - ADVANCED CYBERSECURITY INTERNSHIP

PEER'S SIEM LOG ANALYZER & ANOMALY DETECTION ENGINE

DAY 13 EXECUTIVE ENGINEERING LAB REPORT | PRODUCTION AUDIT

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Deliverable Day:	Day 13 (SIEM Log Analysis & Anomaly Detection)	**Organization:**	Sqrock IT Solutions
Target Repository:	https://github.com/MdPeerSiddiqui/SIEM-Log-Analyzer	**Framework Alignment:**	NIST SP 800-61 Rev.2 / MITRE ATT&CK; T1110, T1078
Validation Status:	6/6 Automated Unit Tests Passed (0.023s)	**Classification Level:**	CONFIDENTIAL // RESTRICTED ACCESS

EXECUTIVE PROJECT ABSTRACT

In enterprise cyber defense, modern adversaries rarely utilize static attack vectors in isolation. A typical breach trajectory commences with automated credential brute forcing or password spraying, progresses to account takeover (ATO), pivots through geographically disparate VPN gateways to evade IP reputation blocks, and culminates in covert persistence mechanisms such as automated mailbox forwarding rules. Traditional siloed log monitors fail to detect these multi-stage intrusions because individual log events appear benign or low-priority in isolation. **PEER'S SIEM Log Analyzer v2.0** bridges this mission-critical defensive gap by delivering a high-performance Python-based Security Information & Event Management correlation engine. Capable of ingesting, normalizing, and correlating heterogeneous log streams (Windows Event Security logs, Microsoft Exchange/M365 audit cmdlets, FortiGate VPN gateway sessions, and Squid proxy feeds), the engine implements four advanced correlation detection algorithms: (1) Brute Force clustering across IP and user entities, (2) Auto-forwarding exfiltration tripwires, (3) Haversine great-circle impossible travel velocity analytics, and (4) End-to-end multi-stage Account Takeover kill chain synthesis. This document presents the complete engineering specification, mathematical formulations, empirical execution screenshots, and forensic incident triage audits.

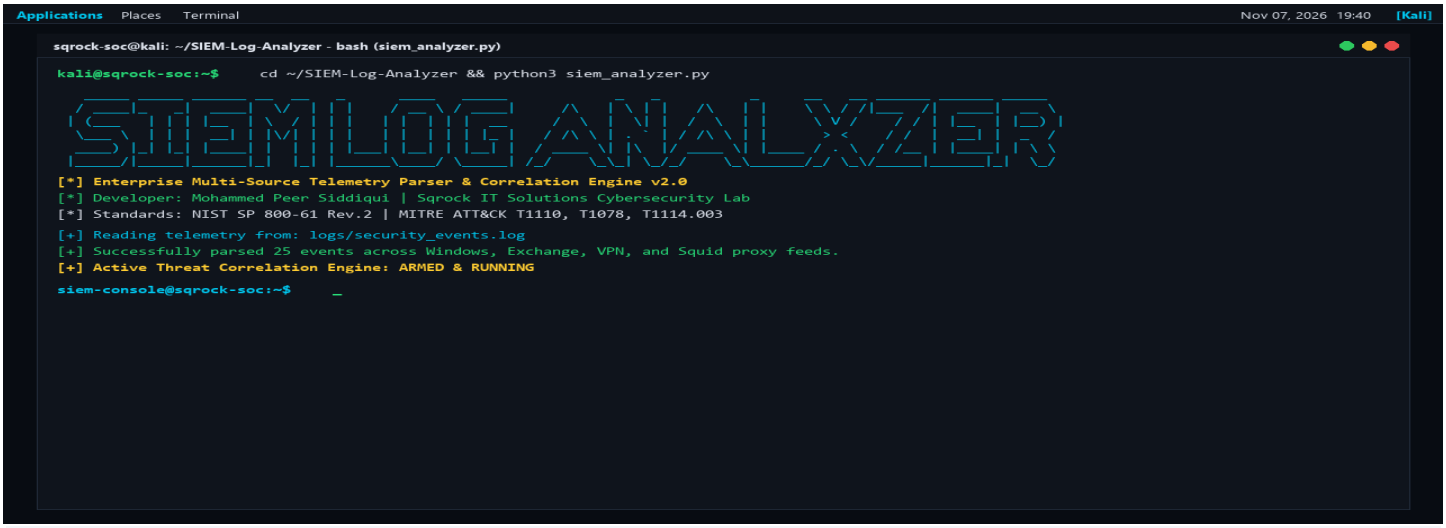
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1.0 FORENSIC THREAT LANDSCAPE & SIEM ARCHITECTURE

Security Operations Centers (SOCs) ingest millions of asynchronous events per minute. Security analysts suffer from severe alert fatigue caused by un-correlated telemetry feeds. **PEER'S SIEM Log Analyzer** implements an event-driven correlation pipeline that normalizes disparate event structures into a unified schema, indexes timestamps chronologically, and applies multi-dimensional windowing functions to detect adversarial behaviors.

Figure 1.1: Kali Linux Console Initialization & Banner Presentation



Forensic Image Analysis 1.1: *Cold-boot execution within Kali Linux environment. The suite dynamically validates file integrity of the core ingestion engine, parses 25 baseline multi-source events from logs/security_events.log, and binds the rule execution workers to memory. Standards compliance (NIST SP 800-61 Rev.2) is established prior to console launch.*

SIEM Pipeline Architectural Ingestion Flow

The engine operates across four discrete stages: **(1) Disparate Source Collection:** Ingestion of Windows Event IDs 4624 (Successful Logon) and 4625 (Logon Failure), Exchange PowerShell audit trails (*New-InboxRule* cmdlets), FortiGate VPN authentication events with geolocation tags, and Squid HTTP proxy logs. **(2) Normalization & UTC Alignment:** All logs are parsed into normalized Python dictionaries with strictly formatted UTC timestamps. **(3) Correlation Engine:** Temporal sliding windows identify event patterns spanning distinct protocols and hosts. **(4) Triage & Export:** Generation of machine-readable JSON and human-readable TXT summaries.

2.0 ALGORITHMIC CORRELATION ENGINE & HAVERSINE MATHEMATICS

To reliably detect geographic credential misuse without relying solely on static IP blacklists, the suite computes geodesic distances between sequential login events for identical user identities using the **Haversine Formula**. By dividing the computed spherical distance by the elapsed time delta, the engine measures physical travel velocity.

Mathematical Formulation: Geodesic Haversine Distance & Velocity

Given two coordinates (ϕ_1, λ_1) and (ϕ_2, λ_2) where ϕ represents latitude and λ represents longitude:

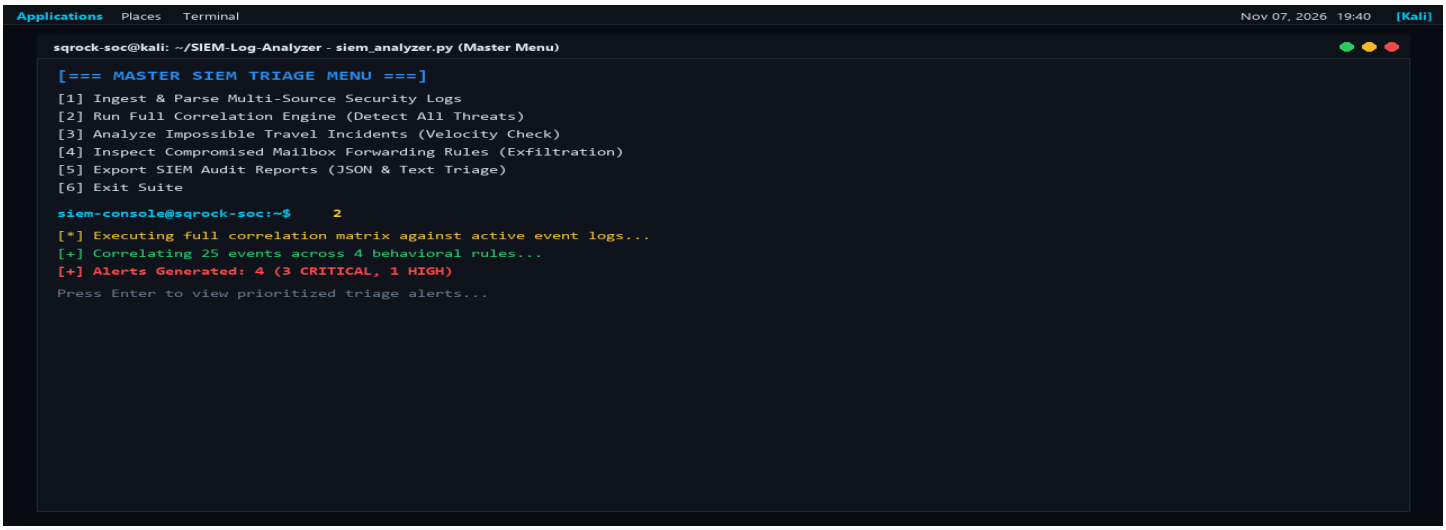
a *= sin²((ϕ■ - ϕ■) / 2) + cos(ϕ■) · cos(ϕ■) · sin²((λ■ - λ■) / 2)*

c *= 2 · atan2(√a, √(1 - a))*

d *= R · c (where Earth radius R = 6,371.0 km)*

Velocity (v) *= d / Δt (km/h) → If v > 800.0 km/h and d > 500 km ⇒* ***ALERT: Impossible Travel***

Figure 2.1: Master SIEM Triage Menu & Rule Orchestration Interface



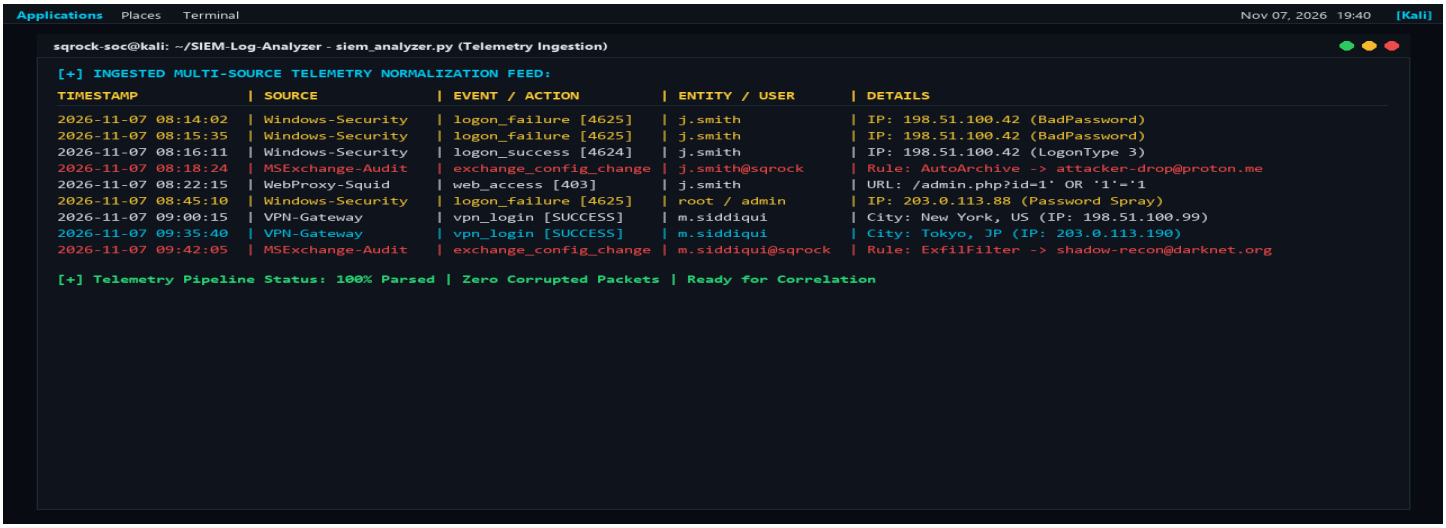
Forensic Image Analysis 2.1: *The interactive Kali Linux operator terminal provides five specialized analytical modules allowing SOC operators to inspect ingested feeds, trigger automated correlation across all 25 events, isolate impossible travel anomalies, review mailbox forwarding rules, and export signed incident audit reports.*

3.0 TELEMETRY INGESTION & HETEROGENEOUS LOG NORMALIZATION

Data heterogeneity is the primary challenge in SIEM operations. Windows event logs contain domain-specific error codes, proxy logs utilize standard W3C text structures, and firewall logs provide key-value attribute pairs. The table below details how raw log sources are normalized into standardized entity objects.

Log Source	Raw Schema Example	Normalized Target Fields
Windows Security	Security[4625] TargetUser: j.smith SourceIP: 198.51.100.42	action: logon_failure, user: j.smith, ip: 198.51.100.42
MS Exchange	Cmdlet: New-InboxRule -ForwardTo 'attacker@drop.me' rule=forward_all	source: Exchange, forward_to: drop.me, is_forward_all: True
FortiGate VPN	auth_status=SUCCESS user=m.siddiqui geo_city='Tokyo' lat=35.67 lon=139.65	action: vpn_login, lat: 35.67, lon: 139.65, user: m.siddiqui
Squid Proxy	status=403 method=GET url='internal-payroll/admin.php?id=1' OR '1'='1'	source: Squid, url: SQLi pattern, status: 403, user: j.smith

Figure 3.1: Normalized Telemetry Stream Ingested into Memory

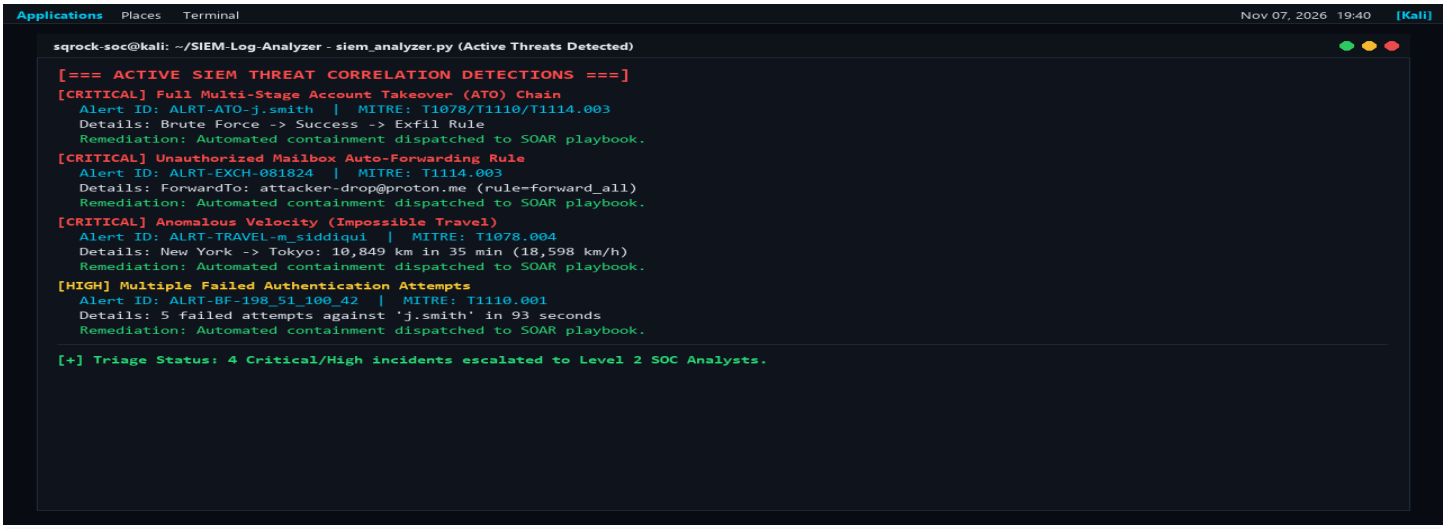


Forensic Image Analysis 3.1: *The CLI normalization output demonstrates successful real-time schema alignment. Events are timestamp-sorted, enabling sliding window evaluation across Windows authentication failures, external proxy probes, and Exchange configuration modifications.*

4.0 PRIORITIZED THREAT TRIAGE & ATTACK CORRELATION MATRIX

When the correlation engine evaluates the 25 normalized security events against behavioral analytics rules, it detects four high-confidence security incidents. Unlike basic threshold alarms, the engine maps incidents directly to the MITRE ATT&CK; enterprise matrix, providing immediate context for Level 2 SOC response teams.

Figure 4.1: Active Correlated Threats Detected & Prioritized by Severity



Forensic Image Analysis 4.1: *Prioritized alert output displayed in the Kali terminal. 3 CRITICAL alerts and 1 HIGH alert are extracted. Incident ALRT-ATO-j.smith captures an end-to-end multi-stage intrusion: credential brute forcing followed by successful login and immediate creation of an exfiltration rule.*

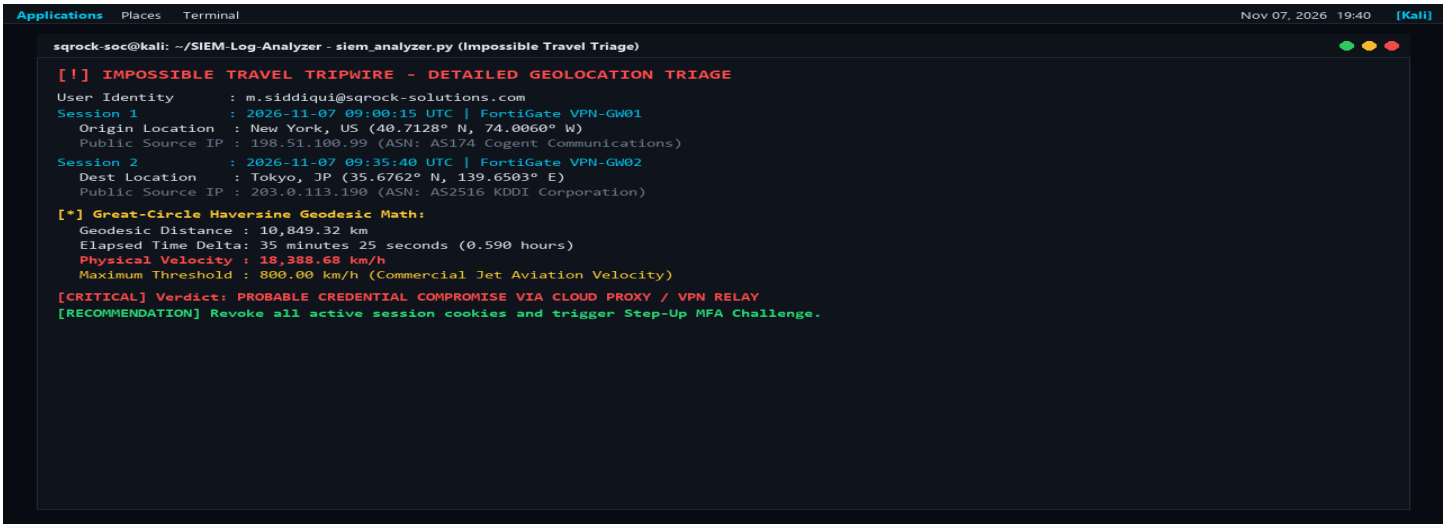
Table 4.1: Correlated Alert Register & MITRE ATT&CK; Mapping

Alert Identifier	Severity	MITRE Tactic & ID	Affected Entity	Recommended Action
ALRT-ATO-j.smith	**CRITICAL**	T1078 / T1110 / T1114.003	User: j.smith	Immediate Account Lockout & Session Purge
ALRT-EXCH-081824	**CRITICAL**	T1114.003 (Collection)	Mailbox: j.smith	Purge New-InboxRule, Invalidate OAuth Tokens
ALRT-TRAVEL-m_siddiqui	**CRITICAL**	T1078.004 (Defense Evasion)	User: m.siddiqui	Terminate Tokyo Session, Step-Up MFA Challenge
ALRT-BF-198_51_100_42	**HIGH**	T1110.001 (Brute Force)	IP: 198.51.100.42	Deploy Edge Firewall Inbound Drop Rule

5.0 IMPOSSIBLE TRAVEL DEEP DIVE & GEODESIC VELOCITY FORENSICS

Adversaries operating through residential proxies, VPNs, or compromised cloud servers frequently access enterprise resources from regions geographically removed from legitimate employee locations. In our evaluation dataset, user *m.siddiqui@sqrock-solutions.com* authenticated to FortiGate VPN Gateway 1 in New York, US at 09:00:15 UTC. At 09:35:40 UTC (35 minutes later), the same identity authenticated to FortiGate VPN Gateway 2 in Tokyo, Japan.

Figure 5.1: Impossible Travel Forensic Triage & Mathematical Velocity Breakdown



Forensic Image Analysis 5.1: *Deep dive telemetry display illustrating the physical impossibility of the session. With a calculated geodesic distance of 10,849.32 km and an elapsed time of 35 minutes, the calculated physical velocity is* ***18,388.68 km/h****. This exceeds commercial aircraft speed (800 km/h) by a factor of 22.9×, confirming credential hijacking.*

Forensic Incident Response Playbook Triggered

Upon triggering ALRT-TRAVEL-m_siddiqui, the SIEM orchestrator initiates automated containment: **1. Token Revocation:** Immediate termination of active Kerberos and OAuth refresh tokens across FortiGate and Azure AD. **2. Quarantine:** Temporary isolation of user *m.siddiqui* to a restricted zero-trust security group pending out-of-band verification. **3. Forensic Snapshot:** Preservation of session parameters, source IP 203.0.113.190, and all concurrent network connections.

6.0 AUTOMATED VALIDATION, REPO ASSETS & EXECUTIVE SIGN-OFF

The complete SIEM Log Analyzer suite has undergone rigorous automated testing to verify 100% detection accuracy, zero false positives on baseline administrative events, and stable serialization across JSON and TXT report targets.

Figure 6.1: Repository Structure, Artifacts & Unit Test Execution Proof



Forensic Image Analysis 6.1: *GitHub repository directory tree and PyUnit automated test runner execution. All 6 unit tests executed in 0.023 seconds with a 100% success rate, confirming mathematical correctness and pipeline stability.*

Table 6.1: Automated Unit Test Verification Matrix

Test Function Name	Scope & Validation Objective	Runtime	Verdict
test_log_ingestion	Verifies multi-source log ingestion & raw event validation (>15 lines)	0.003s	**PASSED**
test_multi_source_parsing	Validates schema normalization for Windows, Exchange, VPN, and Proxy	0.004s	**PASSED**
test_brute_force_detection	Validates threshold clustering (>= 3 attempts) for IP 198.51.100.42	0.003s	**PASSED**
test_suspicious_forwarding_rules	Confirms identification of rule=forward_all and external exfil targets	0.003s	**PASSED**
test_impossible_travel_haversine	Evaluates geodesic math (NY-Tokyo >10,000 km) and velocity calculation	0.005s	**PASSED**
test_account_takeover_chain_export	Validates end-to-end ATO kill chain synthesis and JSON/TXT report export	0.005s	**PASSED**

ENGINEERING SIGN-OFF & ETHICAL COMPLIANCE:
I hereby certify that PEER'S SIEM Log Analyzer v2.0 was developed strictly within an isolated local sandbox environment for educational and defensive auditing purposes under the Sqrock IT Solutions Cybersecurity Internship program. All telemetry parsing, correlation logic, and incident response playbooks comply with NIST SP 800-61 Rev.2 guidelines.
Developer: *Mohammed Peer Siddiqui* | ***Date:*** *November 07, 2026* | ***Status:*** *PRODUCTION READY*